

AALIYAN MANSOOR

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PROFESSIONAL SUMMARY

Computer Systems Engineer, full-stack developer (React/Tailwind), and published AI/Vision researcher. IELTS 8.0 (CEFR C1) certified communicator with a proven ability to architect scalable B2B platforms, deploy edge ML models, and translate complex technical workflows into crystal-clear, publication-ready documentation.

EDUCATION

Bachelor of Engineering in Computer & Information Systems Engineering

Oct 2021 – Sept 2025

NED University of Engineering & Technology, Karachi

SKILLS

Technical Writing	Academic Writing, Technical Documentation, Whitepapers, SEO Content
Frontend & Web	React 19, TypeScript, Tailwind CSS v4, Vite, Node.js, Express, Railway
AI & Hardware	Machine Vision, YOLO, OpenCV, FPGA SoC, Vitis AI

EXPERIENCE

Computer Vision Engineer (in collaboration with Agriauto Industries)

Aug 2024 – Jun 2025

Karachi, Pakistan

- Designed and implemented a real-time machine vision pipeline for steel surface defect detection using YOLO architectures on an FPGA SoC (Avnet Ultra96-V2).
- Managed end-to-end data pipelines, applying advanced dataset augmentation and optimization techniques using Vitis AI to maximize frames per second (FPS) at the edge.
- Authored a comprehensive, highly-commended technical report detailing hardware utilization, architectural efficiency, and inference metrics.

Data Science Intern – National Center of Artificial Intelligence (NCAI)

Oct 2023 – Jan 2024

Karachi, Pakistan

- Collected and preprocessed image datasets of industrial rods for computer vision-based defect detection.
- Applied image processing techniques including thresholding, Canny edge detection, and Laplacian of Gaussian to successfully identify defective components.
- Documented methodologies and translated technical findings into actionable research reports.

PROJECTS

Centaurus Charter — Luxury Yacht Booking Platform

Oct 2024 – Jan 2025

- Engineered a high-conversion, consumer-facing booking platform for a Dubai-based luxury yacht charter.
- Implemented dynamic fleet catalog filtering, internationalization, and optimized Core Web Vitals to maximize global SEO.

AMI CO — B2B Industrial Trading Website

Feb 2026 – Mar 2026

- Architected a full-stack B2B product catalogue and lead generation platform using React 19, Vite 7, and Tailwind CSS v4.
- Handled end-to-end cloud deployment on Railway, DNS management, and Resend email integration.

Global Freight Pakistan & GloPro — Corporate Web Presence

Sep 2025 – Nov 2025

- Engineered robust website deployments and the digital presence for Global Freight and its Portugal-based sister company, GloPro, integrating 3D brand modeling using Spline.

PUBLICATIONS

A. Mansoor, et al. “A Comparative Assessment of YOLO Nano Architectures for High-Speed and Accurate Steel Surface Inspection.” *Advances in Science and Technology Research Journal (ASTRJ)*, Vol. 20, Issue 3, 2026. (Published)

CERTIFICATIONS & ACHIEVEMENTS

- Google Prompting Essentials Specialization (Sep 2025)** – Official 4-course certification mastering advanced LLM prompt engineering and agentic workflows.
- Academic IELTS (Nov 2025)** – Band 8.0 (CEFR Level C1), verifying expert-level English communication for global stakeholder engagement.